

BIOLOGY 101 · ANSWER KEY - PROFESSOR COPY

Homework 2: The Heart and the Blood

Grading guide. Total 24 points. Answers in different words that mean the same thing get full credit; partial credit is suggested where it applies. Enter the final score in Professor mode on the recitation page.

Part A — Label the four rooms

1. For each number on the diagram, write the room's name and what it does (receives blood from where, or pumps blood to where). (8 pts)

Answer: 1 Right atrium — receives used (low-oxygen) blood back from the body. 2 Right ventricle — pumps it to the lungs. 3 Left atrium — receives fresh (oxygen-rich) blood from the lungs. 4 Left ventricle — pumps it out to the whole body. 1 pt per name, 1 per job.

Part B — Harvey's arithmetic, on a newborn

2. A newborn's heart beats 130 times a minute. How many beats in an hour? In a day? Show the multiplication. (3 pts)

Answer: $130 \times 60 = 7,800$ per hour (1 pt). $7,800 \times 24 = 187,200$ per day (1 pt; accept 'about 190,000'). 1 pt for showing the steps.

3. Count your own pulse for 15 seconds (two fingers on the side of your neck or the inside of your wrist). Multiply by 4. What's your heart rate? (1 pt)

Answer: Any reasonable number, usually 70-110 for an 8-year-old. 1 pt for doing it and multiplying by 4.

Part C — What's in the blood

4. Draw a line from each part of the blood to its job. (4 pts)

Answer: Plasma -> The water that carries salt, sugar, and bicarbonate | Red blood cells -> Carry oxygen with hemoglobin | White blood cells -> Hunt and fight germs | Platelets -> Plug leaks — the start of a scab (1 pt each)

Part D — Counting drops

5. A preemie weighs 1.5 kg. About how much blood does she have (80 mL per kg)? A blood test takes 3 mL. What fraction of her blood is that? (Hint: how many 3s fit into her total?) (2 pts)

Answer: $1.5 \times 80 = 120$ mL (1 pt). $120 \div 3 = 40$, so the test is $1/40$ of her blood — about 2.5% (1 pt).

Part E — Flashbacks

6. From NICU Tutor: the umbilical cord has three blood vessels. What are they, and what does a doctor use them for in the first days of life? (2 pts)

Answer: Two arteries (carry the baby's blood out to the placenta) and one vein (brings oxygen-rich blood in) (1 pt). Doctors slide soft catheters into them for live blood pressure, blood samples, and giving fluids or TPN — no skin poke (1 pt).

7. From this morning's Chemistry: the bicarbonate 'sponge' rides in which part of the blood? (1 pt)

Answer: The plasma — the watery part, along with salt, sugar, and proteins.

Part F — Think like a doctor

8. A 900-gram preemie is five days old and breathing harder today than yesterday. The nurse hears a whooshing murmur. An ultrasound of the heart shows the ductus arteriosus is still wide open. Explain what is happening to the blood — and name two things the team might decide to do. (3 pts)

Answer: Blood pumped toward the body is leaking back through the open ductus into the lungs (1 pt), so the lungs are flooded (harder breathing) and the body gets less than its share (1 pt). Options: watch and wait for a small one to close on its own; give a medicine that nudges it shut; rarely, close it with a tiny plug or operation; meanwhile support breathing and go easy on fluids (1 pt for any two).

HOME LAB (optional, not graded): One-way doors in your own arm

You need: your arm, a mirror or a grown-up to watch, and a minute of patience

1. Let one arm hang straight down for a minute until the veins on the back of that hand puff up.
2. Press one puffy vein with a fingertip and keep pressing. With a finger of your other hand, stroke along the vein AWAY from your heart, toward your knuckles.
3. Look at the stretch you just emptied. Does it stay flat, or refill? Now lift the first finger.
4. Raise your hand above your head and watch the veins. Then hang it down again and watch them come back.
5. Bonus: count your pulse for 15 seconds, do 20 jumping jacks, and count again.

What you should see: The emptied stretch stays flat until you lift the pressing finger — a valve (one-way door) stops blood falling back toward the hand. Raised above the head, veins empty because blood runs downhill to the heart; hanging down, they refill. Pulse jumps after jumping jacks. This is Harvey's 1628 demonstration.